

INSTITUT BIOLOGI SISTEM • INSTITUTE OF SYSTEMS BIOLOGY

Dr. Michael H. Gold, MD
Editor-in-Chief
Journal of Cosmetic Dermatology

Dear Dr. Gold,

Please find attached the abstract of our manuscript entitled "Integrated In Vitro and Clinical Evaluation of a Ceramide- and D-Panthenol-Based Moisturizer Demonstrates Enhanced Skin Hydration Through Epidermal Barrier Restoration" for your kind consideration.

Our study presents an integrated evaluation of a ceramide- and D-panthenol-based moisturizer, combining chemical characterization, OECD-compliant non-animal safety assessment, mechanistic investigation using reconstructed human skin models, and clinical efficacy evaluation. We believe the work aligns well with the scope of the Journal of Cosmetic Dermatology.

We would be grateful if you could advise whether you consider this study suitable for publication in the journal. If you find the abstract appropriate and within the journal's scope, we would be pleased to submit the full manuscript for your consideration.

Thank you very much for your time and consideration. We look forward to hearing from you.

Yours sincerely,


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IMAN ILMU AMAL

Integrated In vitro and Clinical Evaluation of a Ceramide- and D-Panthenol-Based Moisturizer Demonstrates Enhanced Skin Hydration Through Epidermal Barrier Restoration

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Abstract

Background: Comprehensive scientific substantiation of cosmetic products increasingly relies on integrated evidence encompassing chemical characterization, non-animal safety assessment, mechanistic validation, and clinical efficacy. However, few moisturizers have been evaluated using such a translational framework.

Aim: This study characterized the bioactive composition of a ceramide- and D-panthenol-based moisturizer and evaluated its antioxidant activity, safety, mechanism of action, and clinical efficacy.

Methods: LC-MS/QTOF was used to characterize the moisturizer, while antioxidant activity was assessed using FRAP, ABTS, and DPPH assays. The safety of the moisturizer was evaluated using OECD-compliant reconstructed human tissue models for skin and eye irritation and skin sensitization. Mechanistic studies in EpiDermFT™ evaluated cytotoxicity, skin barrier gene expression (FLG and CLDN1), IL-8 expression, and histological (H&E and Masson's trichrome staining) analyses. Clinical efficacy was evaluated after four weeks of daily application by measuring skin hydration and transepidermal water loss (TEWL).

Results: LC-MS/QTOF identified D-panthenol together with bioactive compounds associated with skin hydration, tissue repair, antioxidant, and anti-inflammatory activities. The moisturizer exhibited strong antioxidant activity and satisfied all OECD acceptance criteria, confirming the absence of skin and eye irritation and skin sensitization. Mechanistically, the moisturizer was non-cytotoxic, significantly upregulated FLG and CLDN1, downregulated IL-8, and histological analysis demonstrated improved epidermal integrity and tissue organization. Clinical evaluation demonstrated significantly increased skin hydration and reduced TEWL after four weeks, confirming enhanced epidermal hydration and barrier function.

Conclusion: This integrated evaluation demonstrates that the ceramide- and D-panthenol-based moisturizer is safe for repeated topical use and effectively enhances skin hydration by restoring epidermal barrier integrity, reducing inflammatory responses, and providing antioxidant activity. These findings provide robust scientific evidence supporting its application as an evidence-based daily skincare formulation.

Keywords: Ceramide; D-panthenol; moisturizer; skin hydration; skin barrier; non-animal testing.